

Basic Info		
Licensing	Proprietary Cudasip Licence	
Target FPGA or ASIC	ASIC, Demonstrated on VCU118	
Design Language	CodAL3	
System-Level Features		
Coupling Type	Tightly-coupled to A730 Core	
Data Types	INT8,INT16,INT32,INT64,FP32,FP64	
ISA	RISC-V (RV64IMAFDCV)	
Peak Ops Per Cycle	16	
Interfaces	AXI5	
Compiler Support	Y (Automatically generated) or RISC-V compatible compiler	
External Memory Required	No, but supported	
Cache Or Scratchpad	Cache	
Architecture		
Type	Harvard architecture	
Num Clock Domains	1	
Control Interface	ISA Extension	RISC-V (RV64IMAFDCV)
	Memory Mapped Protocol	No AXI5 128b, Custom accelerator
Memory Interface	Interface Protocol	128bit AXI5 (AXI_WIDTH = 2*XLEN)
	Hierarchy Level	Separate L1 cache with shared L2 cache
	Cached	Yes
	IOMMU	Yes
Interrupts	Same as for A730	
Microarchitecture		
Parametrization	Parametric Num Units	One per core
	Parametric Config	No
Programmability	Programmable Cores	Yes (A730)
	ISA Extension	RISC-V (RV64IMAFDCV)
Software		
Compiler	Requires Specialized Compiler Supported Compilers	Yes (automatically generated),or RISC-V compiler compatible with RISC-V V 1.0 gcc, LLVM, Cudasip SDK
highLevelAPI	Has High Level API Has Domain Specific Compiler	Yes - C intrinsics Yes
Integration		
IP Distribution	Manifest Type	None
	Standalone Simulation	No (A730 Required)
	Standalone Simulation SW Requirements	MPC SW Package
	Integration Docs Or Examples	Yes
Synthesis	Is Synthesizable	Yes
	FPGA Synthesis Examples	No
	ASIC Synthesis Examples	No
Simulation	Closed Source Simulation	Yes (QuestaSim)
	Open Source Simulation	No
Evaluation	PPA Results Available	Yes - part of ISOLDE D3.5 deliverable